

INTRODUCTION TO ECONOMICS

MGT404E

Multi-Unit Price Discrimination

Yale

Two ways to price discriminate

Consumers have demand for...

A Single Unit

- Different individuals have different reservation prices
- Cannot necessarily charge everyone their exact reservation price (resale? Arbitrage?)
- Could use observable **demographics** to price.
- Could try to get “high type” consumers to opt-in to paying more. (Self-selecting pricing)

Multiple Units

- An individual values additional units of a product less than they value the first unit
- Ideally charge differently for additional units
- Can try “**membership pricing**” – unlimited units
- Can also try self-selecting:
 - Can offer a **menu** of quantities
 - Can charge a **two-part tariff**.

Now....

Price Discrimination on Quantities

1. Quantity discounting

This is like “perfect” price discrimination for quantities—you simply charge differently for different quantities, to try to capture consumer surplus and DWL as producer surplus.

2. Memberships

In this case, you capture the entire area by setting one price for an “unlimited membership,” which encompasses the varying reservation prices for each unit

3. Quantity Screening

This involves having consumers self-select into quantities of a good. The idea: design a pricing scheme that brings “low types” in while maximizing profits from “high” types.



For this we need to review:

- **Individual demand:** How much an **individual** (*not the entire market*) will demand at a give per-unit price. Reflects an individual’s valuations of multiple units. Individual demand is the cornerstone of multi-unit price discrimination.
- **Valuing a quantity of a good:** The area under an individual demand curve from 0 to q is how much that individual values consuming q units.

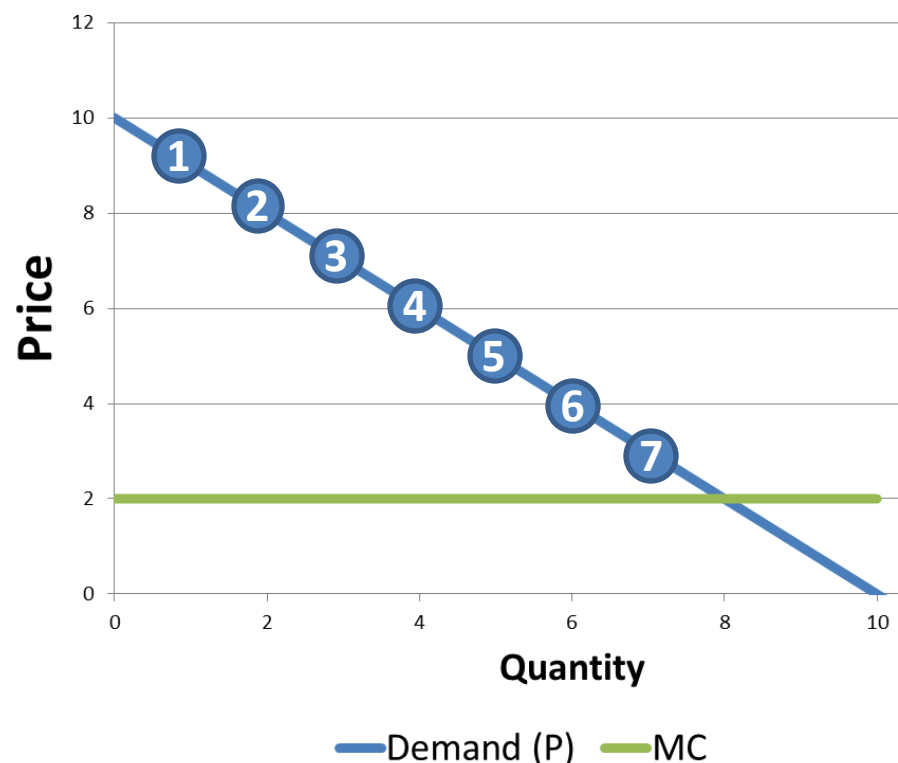
Examples?

1. Quantity discounting
2. Memberships

Quantity Discrimination

Quantity Discrimination

In many settings, an **individual consumer** will value additional units of a good differently. Quantity Discrimination, where a firm sets different prices for different quantities, is another strategy for capturing surplus.



A consumer is willing to pay \$9 for a first unit, \$8 for a 2nd unit, \$7 for a third unit, and so on.

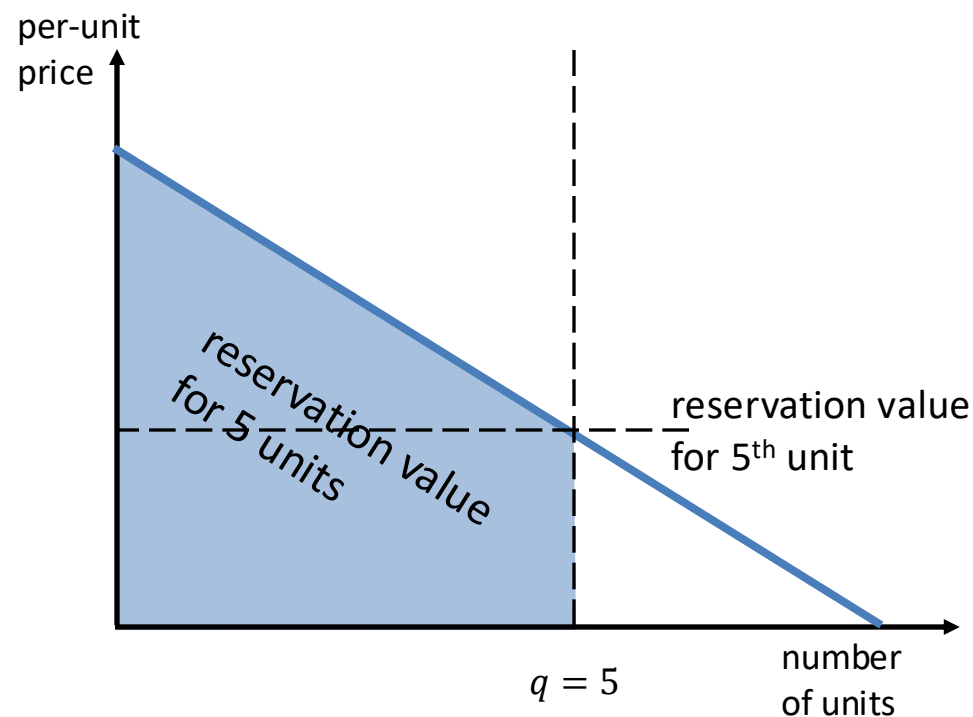
- **Demand curve represents an individual consumer's demand**

Examples:

- Shoes: “Buy one pair, get a pair 50% off”
- Grubhub: discount/free delivery/bonus based on amount ordered
- Note well: we are switching from drawing the MARKET demand to drawing individual demand.

Individual demand

Individual demand is how much one consumer will purchase at a given per-unit price. It is **not** the same as a market demand.

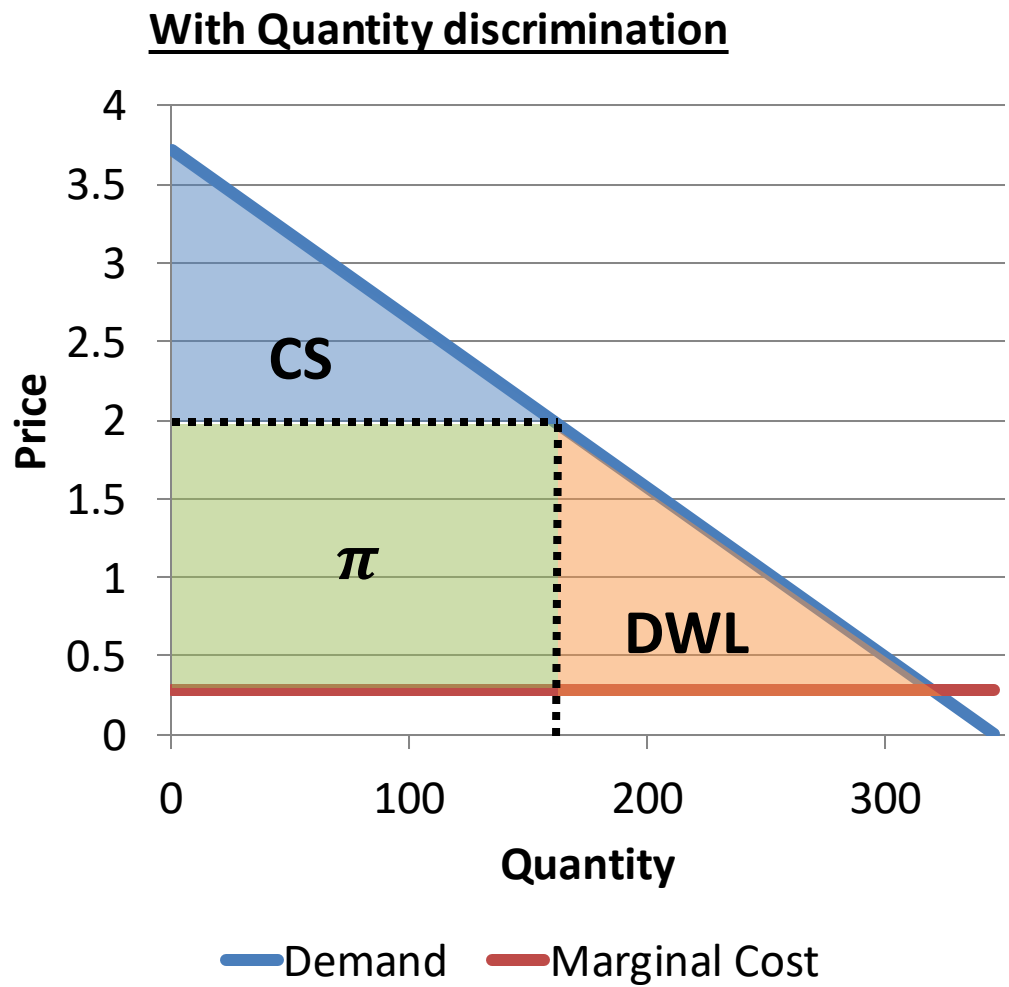


Each point on an individual demand curve represents the value of a particular unit of consumption.

An area under an individual demand curve represents the reservation value for a certain bundle of consumption units.

As with market demands, we usually assume they are lines rather than stair-steps.

Quantity discrimination implications: Same regions, but each curve represents one consumer



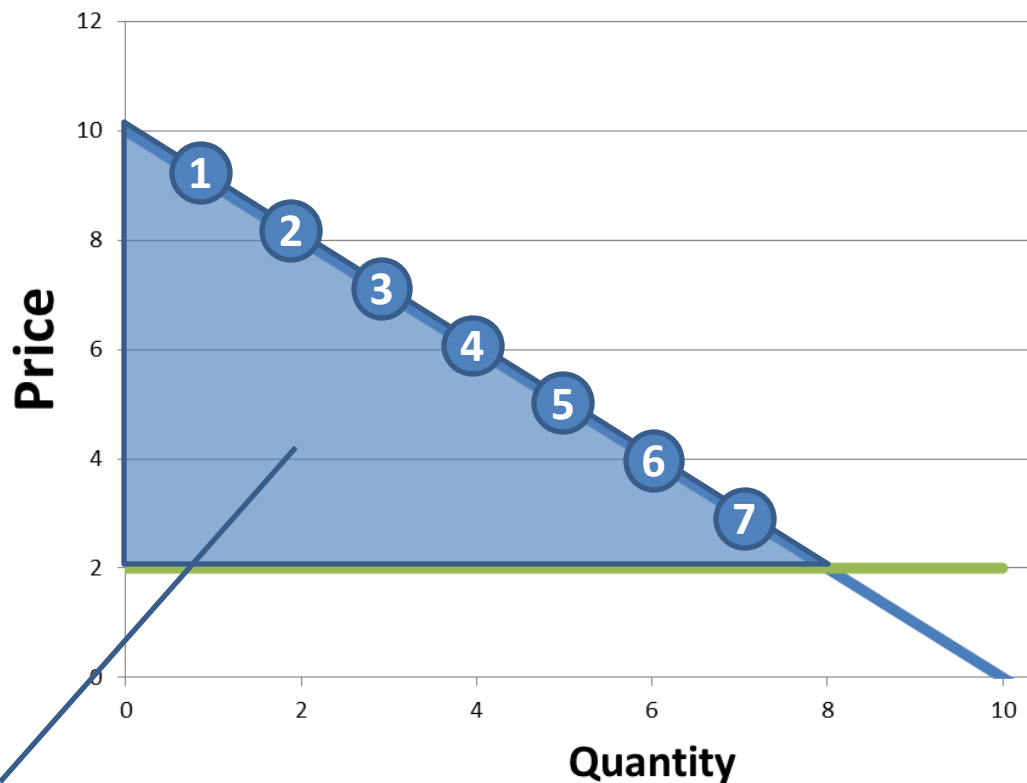
Lower price for subsequent units: Go ahead and provide a discount conditional on purchasing a certain quantity. Gets rid of DWL..

Higher price for earlier units: But the monopolist won't stop there: they should charge a higher price for initial units, and do quantity discrimination all the way down the curve.

How to set prices to capture DWL?

1. Perfect Quantity Discounting

Perfect Quantity discrimination



Firm Profits

— Demand (P) — MC

Same approach as Perfect Price Discrimination:

- The firm knows the consumer's reservation price for each unit
- Charge each consumer exactly their reservation price

Result: Firm captures all possible surplus

- No deadweight loss, efficient outcome
- Again, if re-sale were possible, this would not work.

Examples?

Examples: BOGO1/2 and “Buy more save more”

1/2 Price Dinner

season's best
buy one, get one
50% off
offer good on ALL SIZES

smartcanucks.ca
BUY ONE GET ONE

BUY MORE, SAVE MORE
ONLINE ONLY! Valid through Sunday, May 5, use promo code **SENDTOME**

\$10 OFF
a purchase of \$25 or more

\$15 OFF
a purchase of \$50 or more

\$20 OFF
a purchase of \$100 or more

sale
buy one, get one
50% off
all Vitamin Shoppe® & BodyTech® brand products mix & match

Brooklyn
(732) 9

CS THE GROWLER STATION

BUY 1 GROWLER OR BEAST GET THE 2ND 1/2 OFF
(2ND MUST BE OF EQUAL OR LESSER VALUE)

it's BOGO time!
shop now

EV
Buy 1, get Payless Sh

BUY MORE, SAVE MORE

25% off
WHEN YOU SPEND \$50 OR MORE

20% off
WHEN YOU SPEND \$40 OR MORE

15% off
YOUR ENTIRE ORDER

CouponAlbum.com

Example: Chicken Wing Madness

WINGS

4 Chicken Wings	4.55	24 Chicken Wings	27.25
5 Chicken Wings	5.70	25 Chicken Wings	27.80
6 Chicken Wings	6.80	26 Chicken Wings	28.95
7 Chicken Wings	7.95	27 Chicken Wings	30.10
8 Chicken Wings	9.10	28 Chicken Wings	31.20
9 Chicken Wings	10.20	29 Chicken Wings	32.35
10 Chicken Wings	11.35	30 Chicken Wings	33.50
11 Chicken Wings	12.50	35 Chicken Wings	39.15
12 Chicken Wings	13.60	40 Chicken Wings	44.80
13 Chicken Wings	14.75	45 Chicken Wings	50.50
14 Chicken Wings	15.90	50 Chicken Wings	55.60
15 Chicken Wings	17.00	60 Chicken Wings	67.00
16 Chicken Wings	18.15	70 Chicken Wings	78.30
17 Chicken Wings	19.30	75 Chicken Wings	83.45
18 Chicken Wings	20.40	80 Chicken Wings	89.10
19 Chicken Wings	21.55	90 Chicken Wings	100.45
20 Chicken Wings	22.70	100 Chicken Wings	111.25
21 Chicken Wings	23.80	125 Chicken Wings	139.00
22 Chicken Wings	24.95	150 Chicken Wings	166.85
23 Chicken Wings	26.10	200 Chicken Wings	222.50

Danny's Wok, West Philadelphia

To an economist, this is a “cursed image”

... what were they thinking!?

As you get drawn into the image, you might start to notice things like the price of $50 + 10 < 60$...

Chicken Wing calculus

AVERAGE PRICE OF WINGS



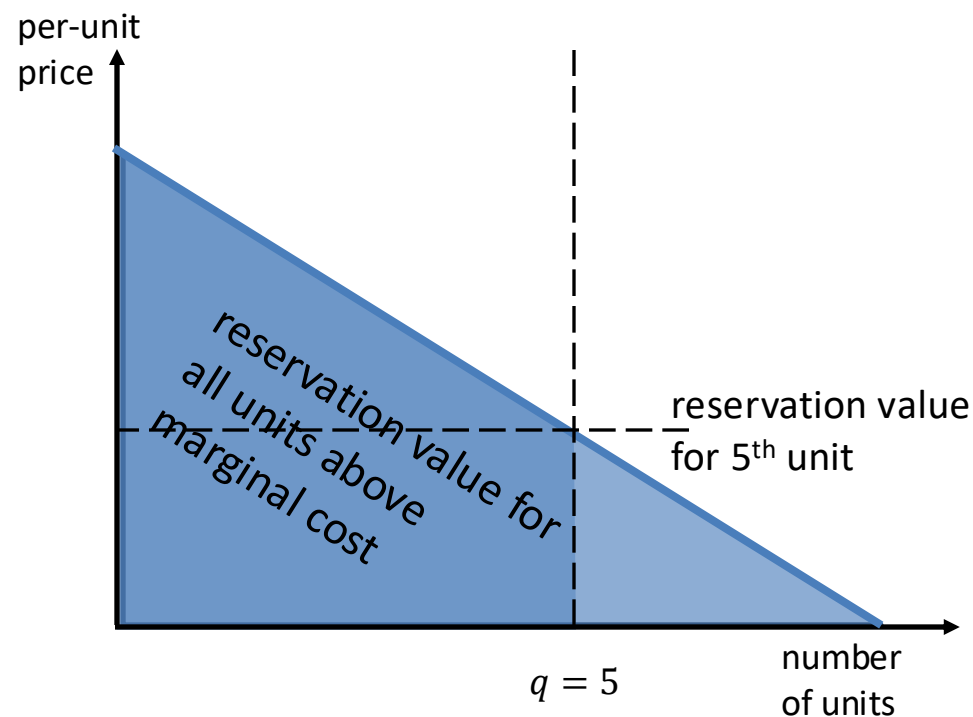
The number of wings divided by total price



Source: The Verge

2. Membership Pricing

Remember individual demand? Since the consumer has a total valuation of a certain quantity of units, you can also charge them a fixed price for a *package*.



An area under an individual demand curve represents the reservation value for a certain bundle of consumption units.

It is a sum—the highest price for the first unit, plus the next price for the second, etc.

Membership fees

Suppose my reservation price for gym visits in a month is given by $P(q) = 16 - q$

- This implies I would be willing to pay \$15 for my first gym visit this month, \$14 for my second, etc., until I wouldn't be willing to pay anything for my 16th visit of the month.
- The gym, New York Sports Clubs (NYSC), knows this and knows that their MC for each visit is \$0.

Membership fee

What should NYSC charge for my membership?

- Given a membership, I will go 16 times.
- My reservation value for 16 visits is
$$0.5 \cdot 16 \cdot \$16 = \$128$$
- This is what NYSC should charge. It is also the profit they make off of me.

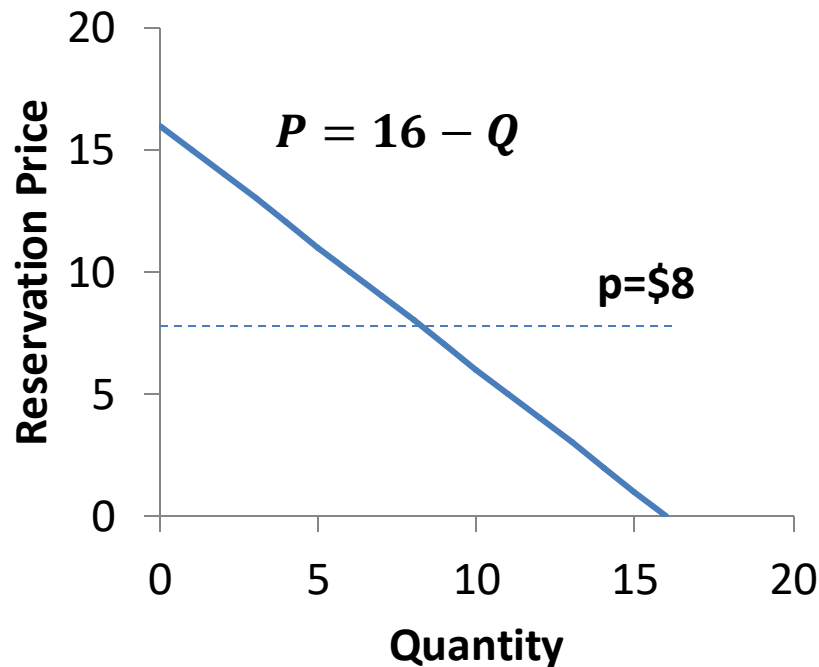
Per-visit fee

What per-visit fee should NYSC charge?

- Their profit from me is
$$\pi(q) = P(q) \cdot q = 16q - q^2$$
- Setting the derivative to zero yields $q^* = 8$, and hence they should charge $p^* = P(8) = \$8$ per visit.
- Profit is only \$64

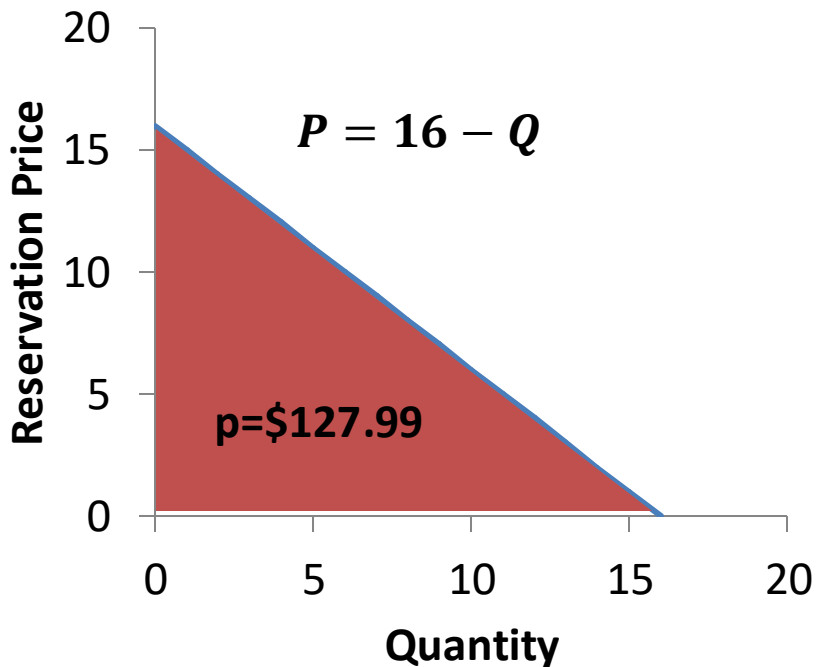
Note: Graphing Uniform vs Membership Prices

Uniform Prices (Price Per Unit)



- Price is a **line** in this graph.

Membership Prices (Total Price)



- Price is an **area** in this graph. It is for ALL units, not a single unit.
- Also applies whenever solving for a price for a specific quantity of the good.

Classic Example of Membership Pricing



MEMBERSHIP OPTIONS

94th & Broadway

Pick the option that's right for you.



Passport

Enjoy unlimited access to all of
our sports clubs*

MONTH-TO-MONTH	1 YEAR COMMITMENT
\$89⁹⁹ /mo	\$79⁹⁹ /mo
JOIN	JOIN

Membership Pricing Cautionary Tale

1981: American Airlines introduces “AAirpass”, unlimited lifetime first class travel for \$250,000. 66 were sold, company was under financial stress.
Price increased to \$600K in 1990, \$1M in 1993.



How is first class air travel different from a gym pass?

For a consumer: what is the marginal price of an additional trip?

Answer: 0

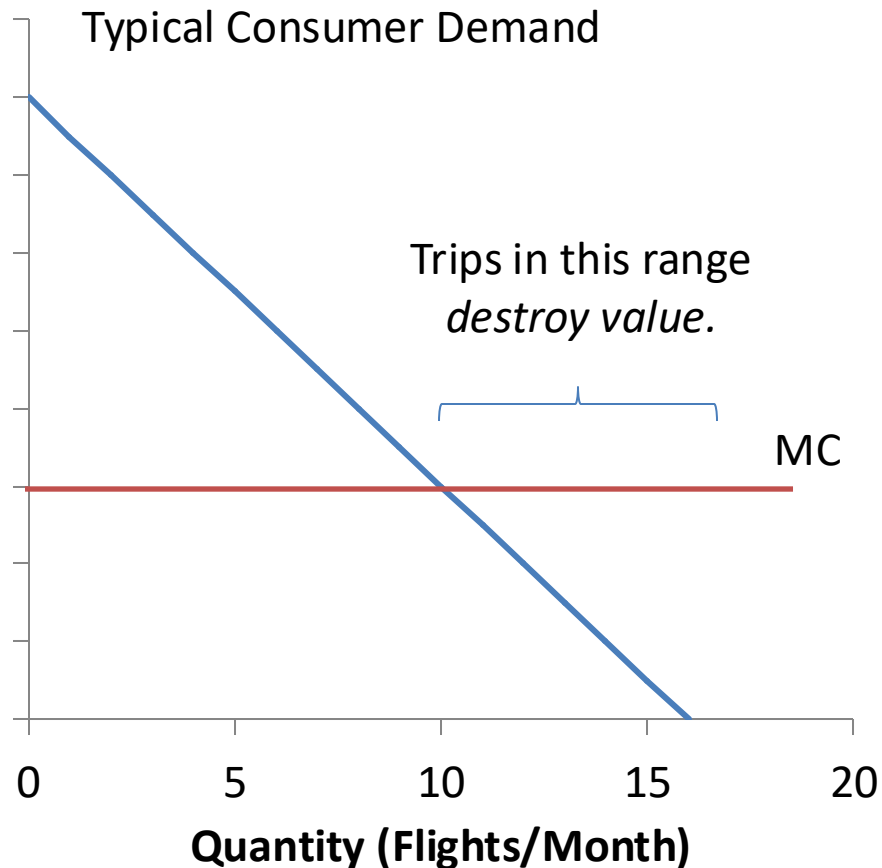
For the firm: what is the marginal cost of an additional passenger-trip?

Answer: more than 0.

Any pricing scheme where marginal price < marginal cost is unlikely to work out.

“It’s all a great selling proposition,” said Robert Mann, an airline industry analyst and former executive who helped American Airlines begin an all-you-can-fly pass in 1981. “The key is very few people make money doing it.”

AAirpass Membership Pricing



Membership pricing has both pros and cons:

Pros:

- Consumers value simplicity
- Easy to implement

Cons:

- Mis-match between *marginal price* and *marginal cost*
- Can lead to value destruction: consumer values good at less than cost, but it is provided anyway.

Pure membership pricing works best when marginal cost is near zero.

Quantity Screening

For these strategies, different valuation by consumers as well as by quantity

Consumers demand multiple quantities of a good. However, there are also different types of consumers—it's impossible to know each of their reservation values perfectly. So, instead, we design packages they will want to self-select. But here, they differentiate themselves on valuation of quantities, rather than other characteristics.

Strategy

Notes

Menu Pricing

- For many goods, it is not feasible to allow customers to choose any quantity.
- This strategy offers quantity bundles targeted at different consumer types, which consumers may self-select into.

Two-Part Tariffs

- Charging a “membership” fee and a “usage” fee allows a firm to target different consumer segments, and allow them to select any quantity they would like.

Menu Pricing Example



1 GB
4G LTE DATA

\$50 /MO. INCLUDES:
UNLIMITED TALK,
TEXT & DATA
[More Details >](#)

3 GB
4G LTE DATA

\$60 /MO. INCLUDES:
UNLIMITED TALK,
TEXT & DATA
[More Details >](#)

5 GB
4G LTE DATA

\$70 /MO. INCLUDES:
UNLIMITED TALK,
TEXT & DATA
[More Details >](#)

UNLIMITED
4G LTE DATA

+ Rhapsody unRadio
subscription included

\$80 /MO. INCLUDES:
UNLIMITED TALK,
TEXT & DATA
[More Details >](#)



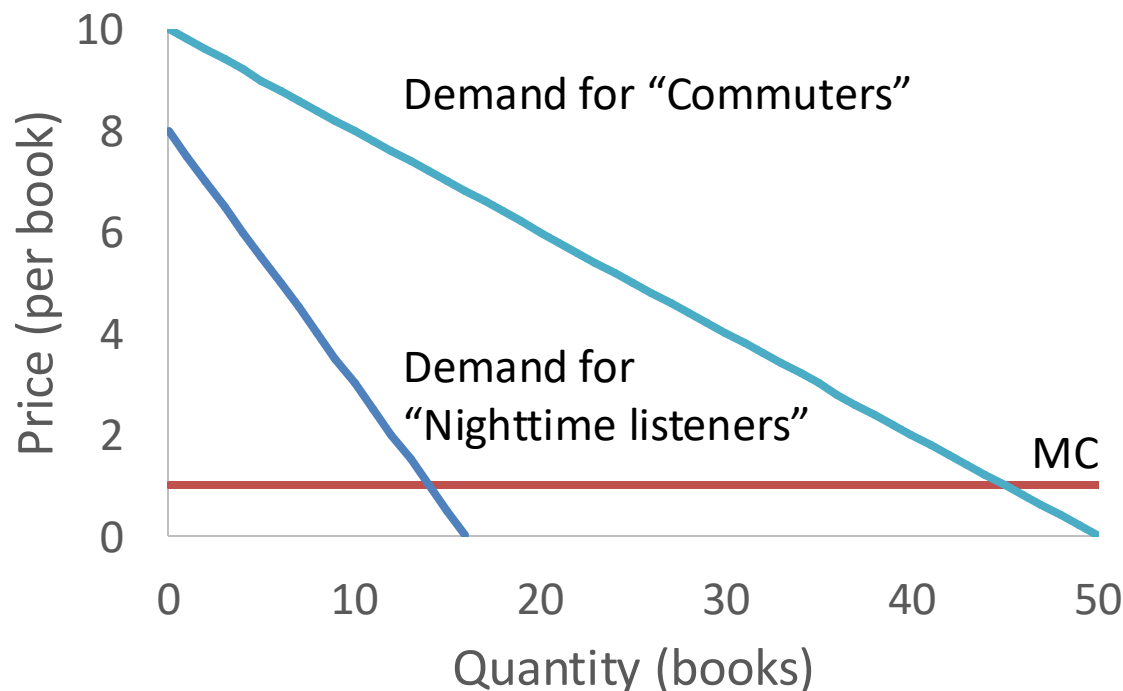
- Note these are essentially different memberships with different “cutoff” points.
- The memberships use the principles of quantity discounting, while different levels use the principles of self-selecting price discrimination.
- How should you design these cutoff points optimally?

Menu Pricing

Example: Audible Memberships. Constant marginal cost per book. of \$1. Two types of consumers: “Nighttime listeners” and “Commuters”. Each type can buy one of two memberships options. Yearly book demand is:

$$P_n(q) = 8 - 0.5q$$

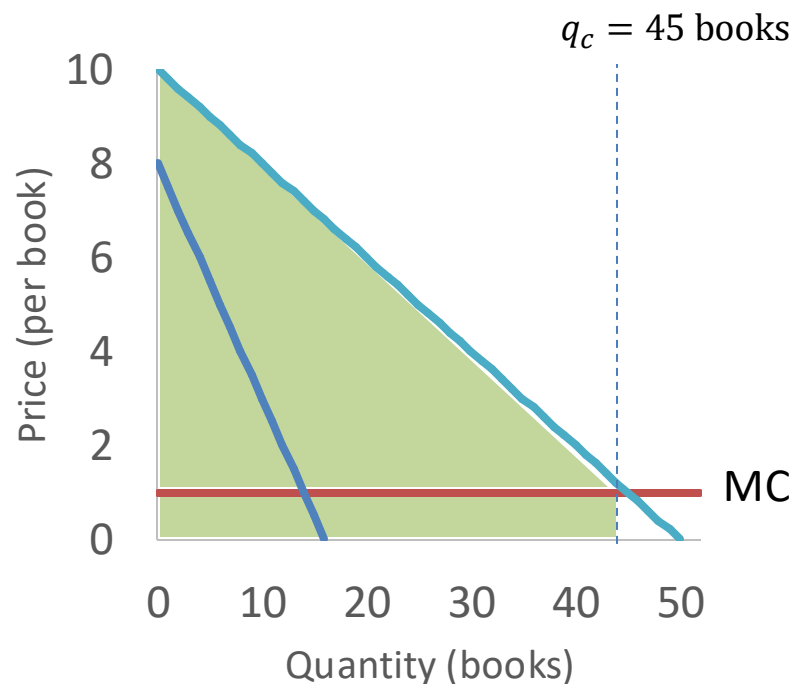
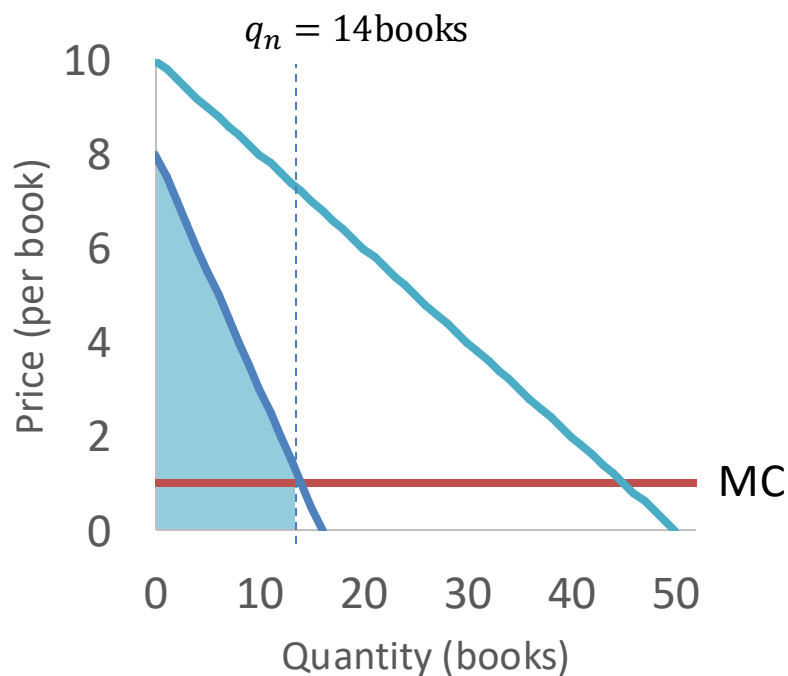
$$P_c(q) = 10 - 0.2q$$



- Nighttime have a high value on the first few audio books, then demand falls off
- Commuters have higher values for all books
- It is not feasible to let consumers choose an exact quantity in books.
- What membership sizes (in books per year) should we offer?

Menu pricing with perfect price discrimination

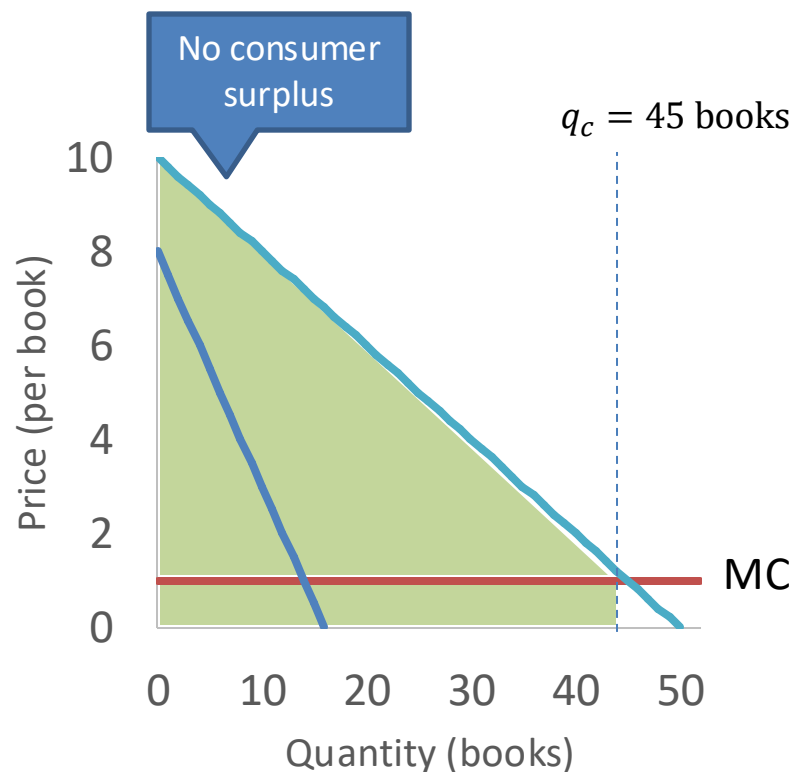
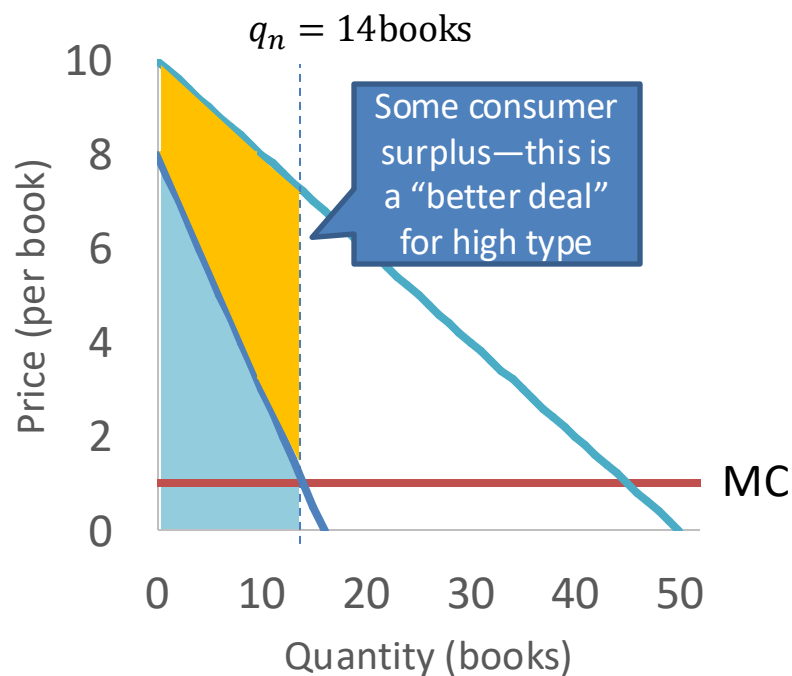
If we could tell the two types apart, we would use our membership pricing rules: biggest size possible (where demand meets MC), for the biggest price (CS).



Question: If we tried this pricing approach, would the “Commuters” get more consumer surplus from buying the “small” or the “large” subscription?

Menu pricing without perfect price discrimination

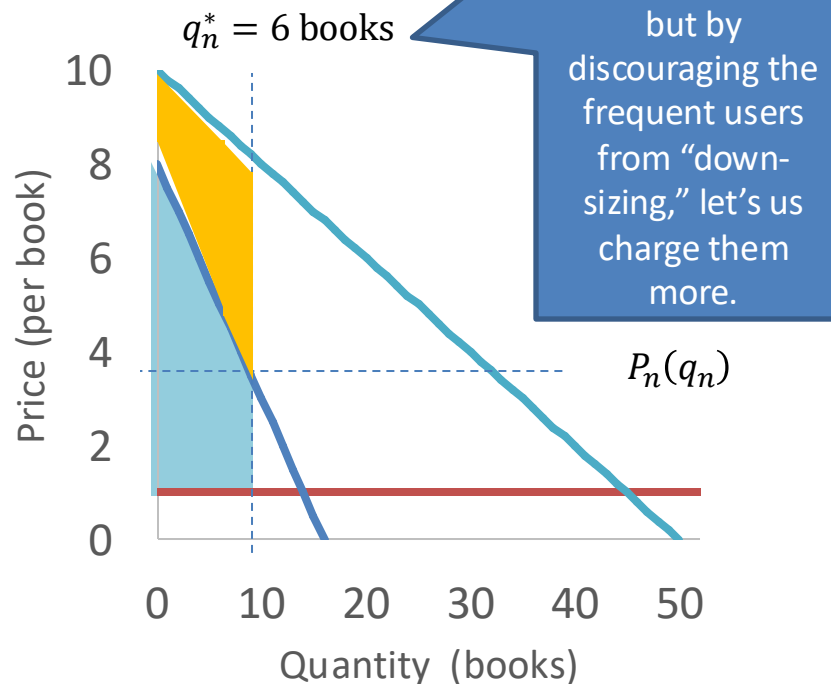
But if we cannot tell the two types apart, Nighttime listeners would buy the small membership, but what about commuters?



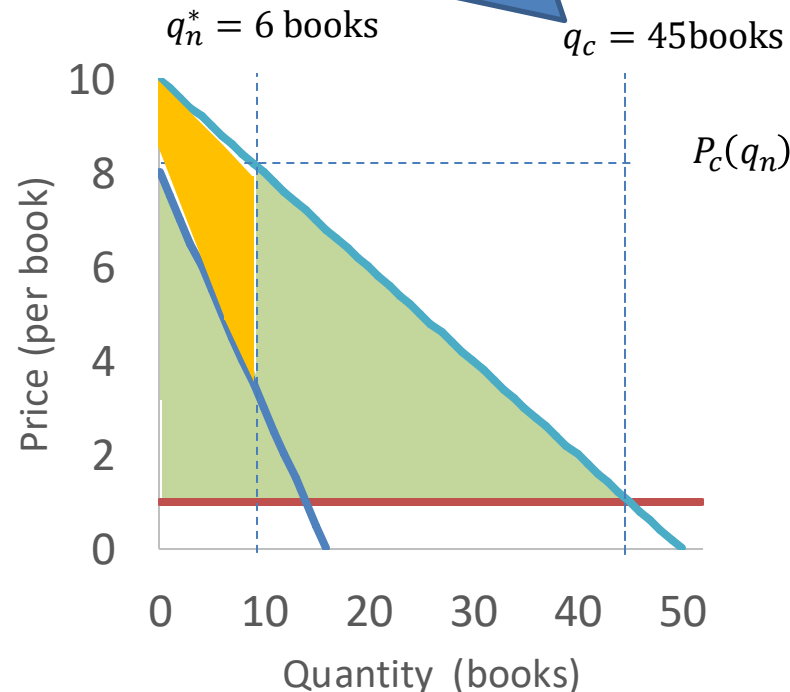
Need to leave as much consumer surplus on Commuter membership as they would get by purchasing Nighttime membership—then they'll purchase the larger package, and you'll earn more.

Menu pricing: tradeoffs

Let's try a smaller size.



We never make the large smaller, because keeping it big gives us more CS to “work with”



How much smaller we want to make the small depends on how many “low demand” types there are compared to high types.

- If more high, you care more about lost profit on high pricing, so make small package even smaller to keep them away.
- If more “low,” you make the small bigger. Implication? As # of nighttime in population increases, Audible makes is small package *larger*.

How may we help?



[Help Center](#) > What are the benefits to signing up for an Audible membership?

What are the benefits to signing up for an Audible membership?

Being enrolled in an Audible membership gives you access to a number of exclusive benefits and promotional offers that save you money and improve your listening experience. Membership benefits vary by membership plan.

	Plans					
	Gold Monthly	Platinum Monthly	Gold Annual	Platinum Annual	Audible Channels	Amazon Prime
Price	\$14.95 per month	\$22.95 per month	\$149.50 per year	\$229.50 per year	\$4.95 per month	See pricing at Amazon.com/Prime
Number of Credits	1	2	12	24	None	None
Frequency of Credits	Monthly	Monthly	Annually	Annually	N/A	N/A
Rollover Credits	5 credits per month	10 credits per month	6 credits per year	12 credits per year	N/A	N/A

Related Answers

How can I set up my complimentary subscription of Wall Street Journal and New York Times?

How can I cancel my complimentary subscription?

What happens if I do not use my monthly or yearly Audible credits?

What is the Great Listen Guarantee?

What happens to my audiobooks if I cancel my membership?

Quantity Self-Selecting Price Discrimination

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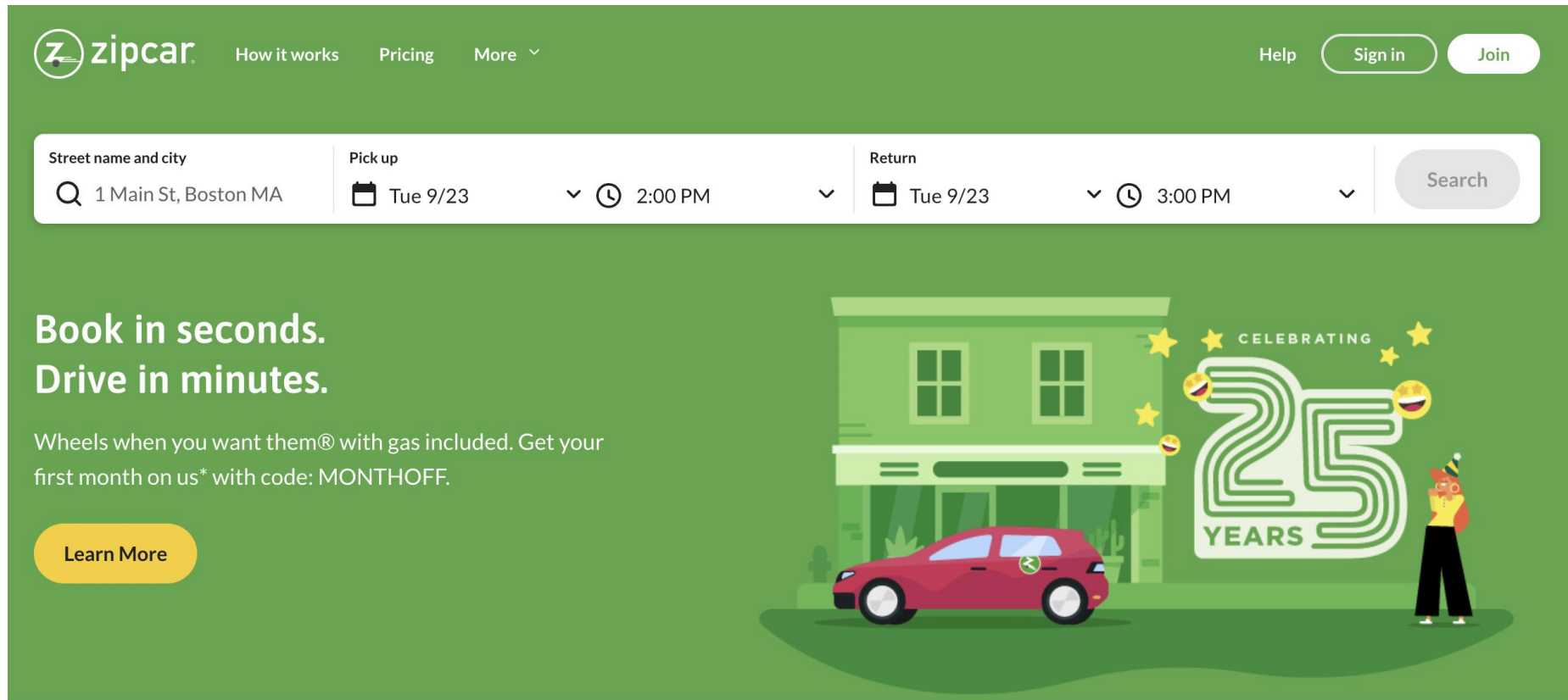
Menu Pricing

- For many goods, it is not feasible to allow customers to choose any quantity.
- This strategy offers quantity bundles targeted at different consumer types, which consumers may self-select into.
- Same logic as versioning

Two-Part Tariffs

- Charging a “membership” fee and a “usage” fee allows a firm to target different consumer segments, and allow them to select any quantity they would like.

Motivation: ZipCar



- Car sharing service launched in 2000
- Some customers have higher demand for the service than others: want to design a pricing plan that appeals to all types of consumers.
- Solution: a membership fee in addition to an hourly usage fee

Examples of Two-Part Tariffs

Definition: Two-Part Tariff

A two-part tariff is a pricing plan where consumers must first pay a lump-sum fee for the right to purchase (the first tariff), and **then a per-unit fee for each unit purchased** (the second tariff, or the “usage fee”)

Example

Description

Private Golf Course

- Members pay an “initiation fee” (tariff), and then green fees every time they golf (usage fee).

Zipcar

- Members pay a monthly or annual membership fee (tariff), as well an hourly rate (usage fee).

County Fair

- Guests pay for admission to the fairgrounds (tariff), but then also pay for individual rides (usage fee)

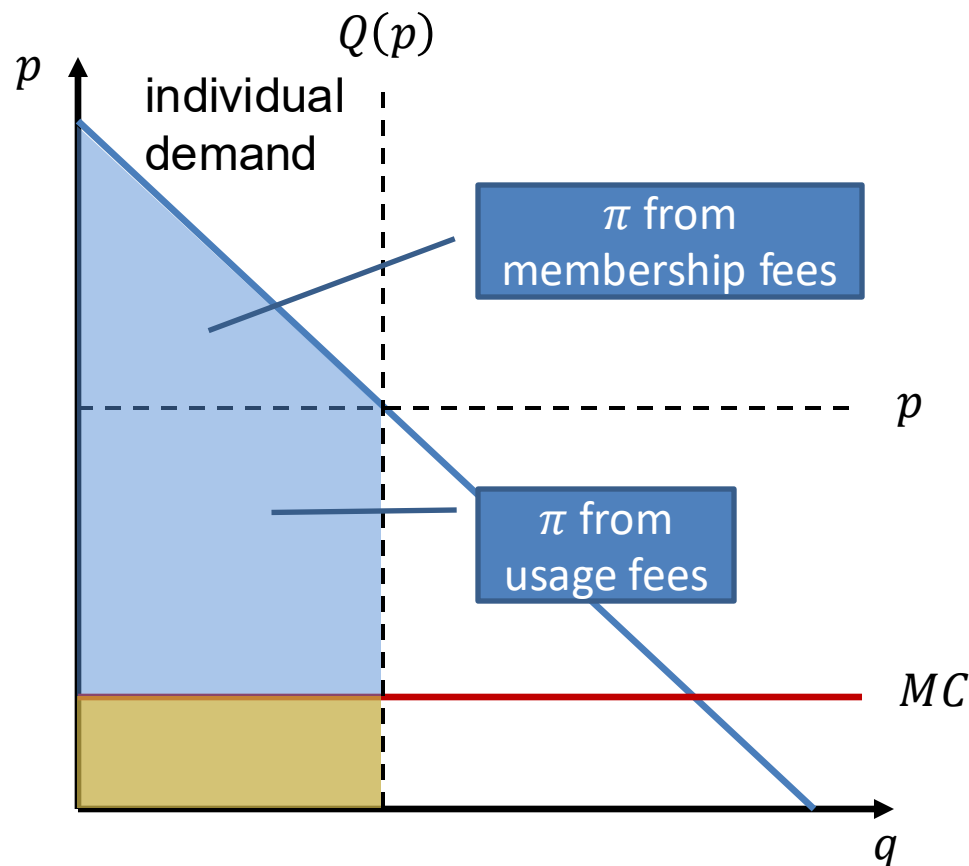
Two-Part Tariffs : Identical Consumers

The Situation

Zipcar is trying to determine the optimal two-part tariff for its service.

- Potential consumers in the market are identical, each with annual demand for hours of car rental given by $P = 38 - Q$
- Marginal cost to the company for an hour of car rental is \$2.

What usage fee and membership fee should they set for a two-part tariff?



Optimal Two-Part Tariff with Single Consumer Type

What's the maximum membership I can charge for any price?

- The remaining consumer surplus
- Can I do better? Is this the correct price?

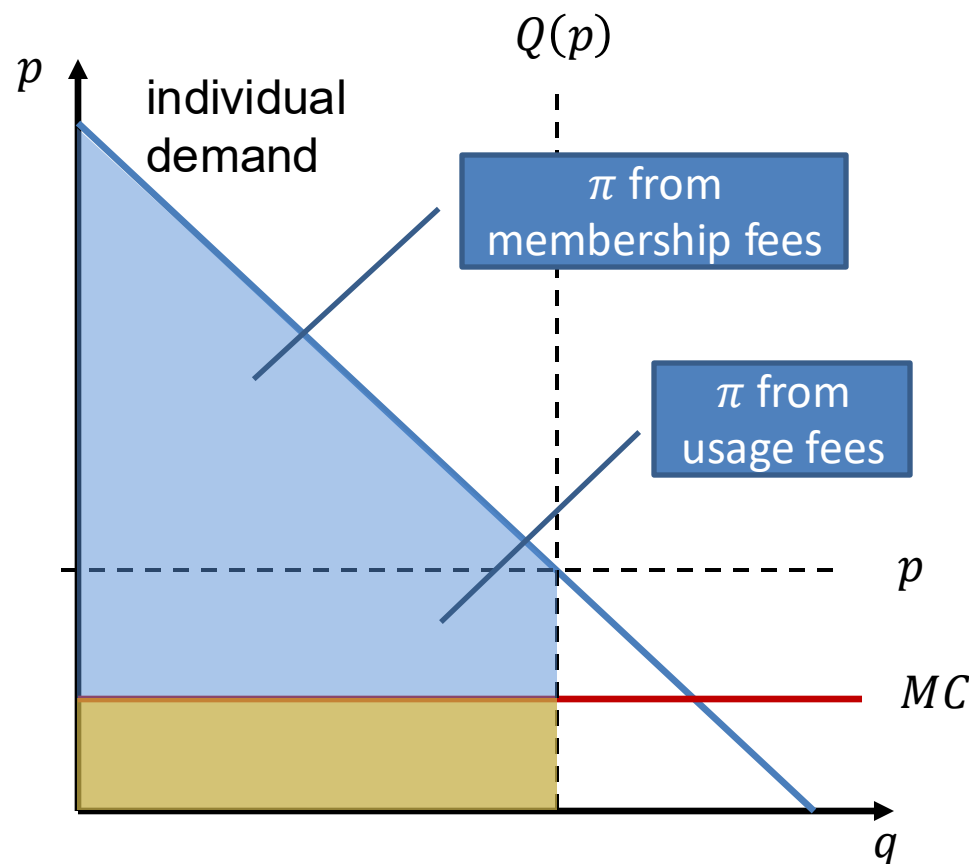
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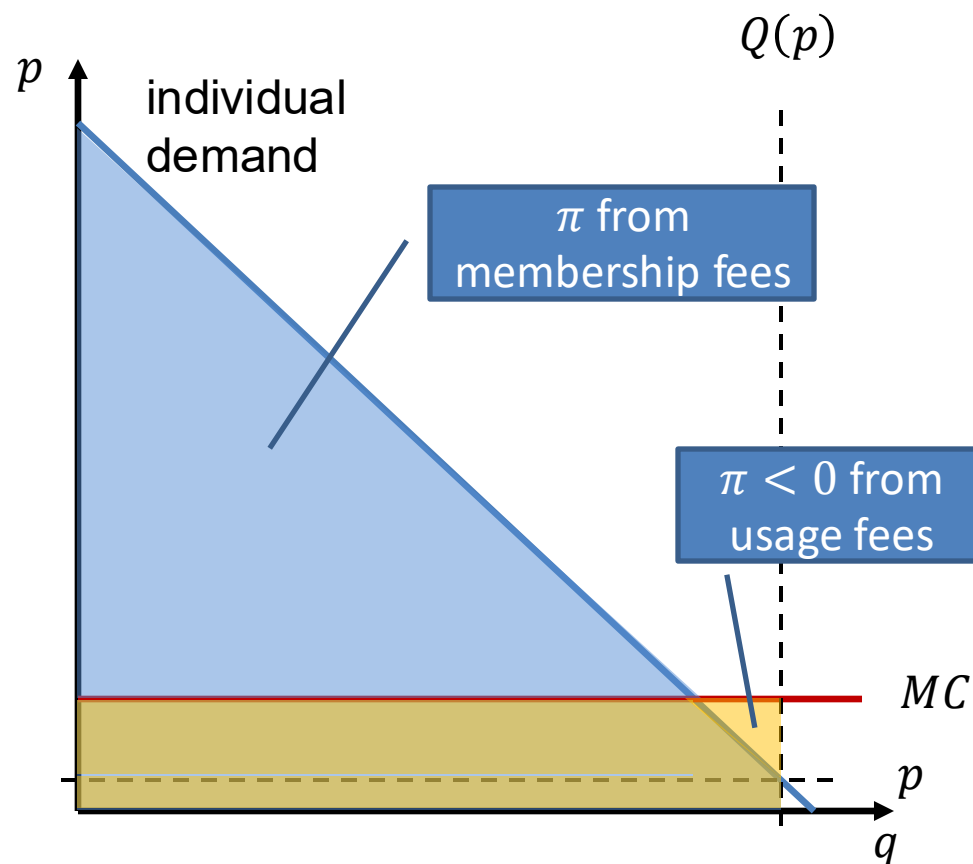
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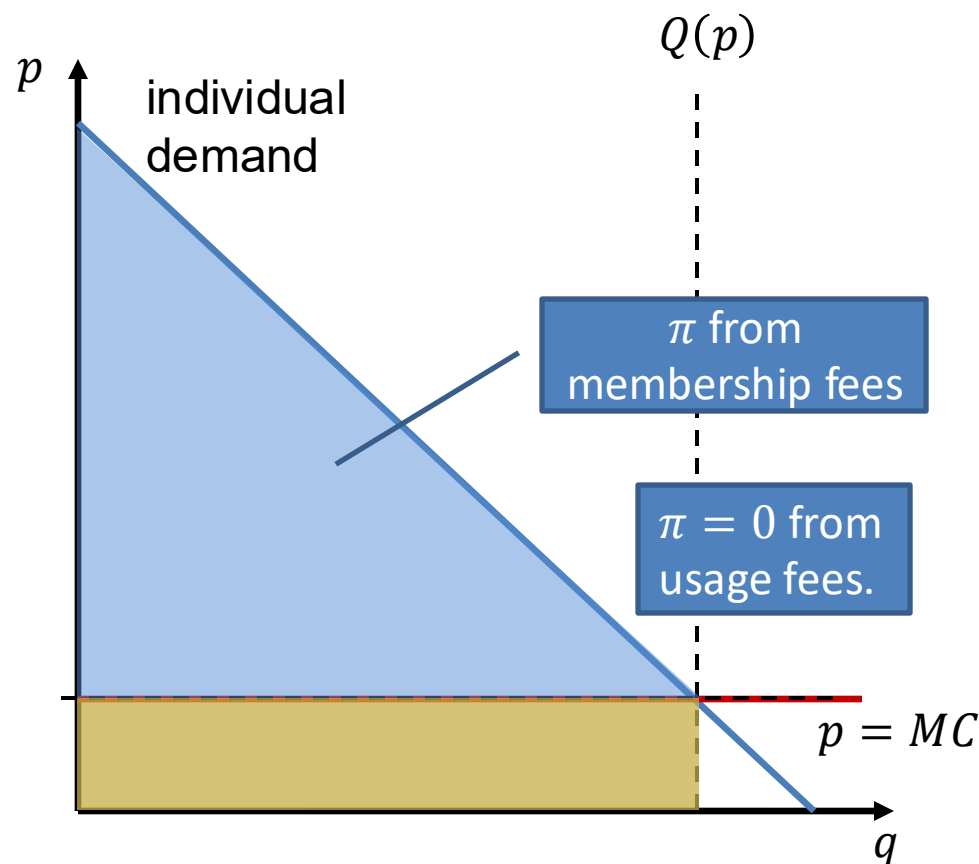
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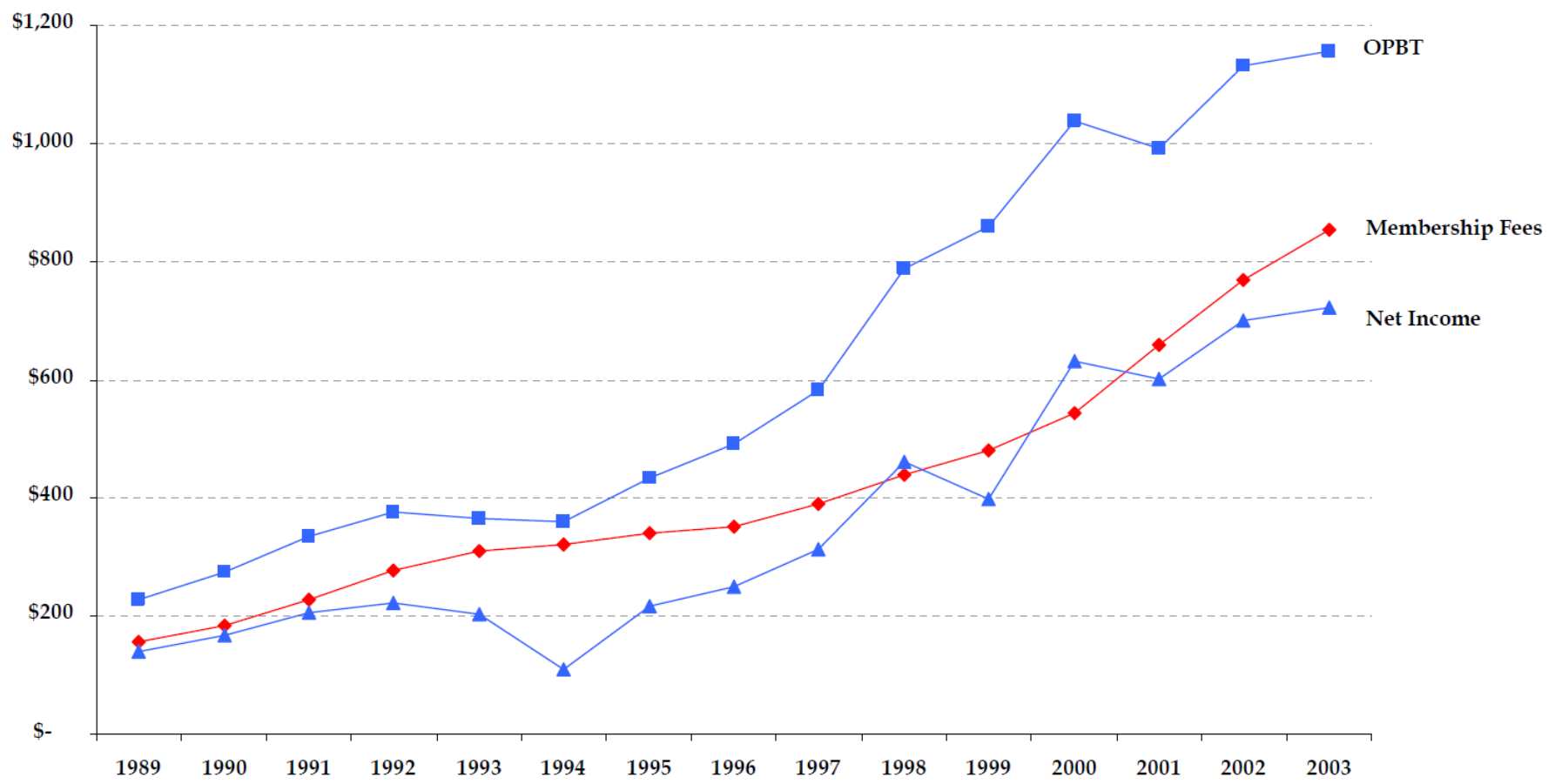


Optimal Two-Part Tariff with Single Consumer Type

Set usage fee to be marginal cost; set membership fee to be the consumer surplus given the usage fee. With a single type of consumer, a two-part tariff allows the firm to extract all surplus, resulting in greater profits than a monopolist using uniform pricing

In Action: Costco

Costco OPBT, net income and membership fees by year
\$, millions, FY1989-FY2003



Costco model: membership fees make up most of profit



COSTCO WHOLESALE CORPORATION
CONSOLIDATED STATEMENTS OF INCOME
(amounts in millions, except per share data)

	52 Weeks Ended September 1, 2013	53 Weeks Ended September 2, 2012	52 Weeks Ended August 28, 2011
REVENUE			
Net sales	\$102,870	\$ 97,062	\$ 87,048
Membership fees	2,286	2,075	1,867
Total revenue	105,156	99,137	88,915
OPERATING EXPENSES			
Merchandise costs	91,948	86,823	77,739
Selling, general and administrative	10,104	9,518	8,691
Preopening expenses	51	37	46
Operating income	3,053	2,759	2,439
OTHER INCOME (EXPENSE)			
Interest expense	(99)	(95)	(116)
Interest income and other, net	97	103	60
INCOME BEFORE INCOME TAXES	3,051	2,767	2,383
Provision for income taxes	990	1,000	841
Net income including noncontrolling interests	2,061	1,767	1,542
Net income attributable to noncontrolling interests	(22)	(58)	(80)
NET INCOME ATTRIBUTABLE TO COSTCO	\$ 2,039	\$ 1,709	\$ 1,462

Possible Application (3): Keurig and K-cup



- Keurig adopted a two-part tariff:
 - Charge for the machine
 - Charge for usage (k-cups)
- How much should Keurig charge for machine?
How much for k-cups?

Possible Application (3): Keurig and K-cup



- What if some consumers just want to enjoy an occasional cup of coffee?
 - They may not want to pay for expensive machine
- What is the solution?
 - **Lower price of machine;**
 - **Charge usage fee > MC.**
- What do you think happened?

COFFEE WITH PURPOSE



Possible Application (3): Keurig and K-cup



- What is the solution?
 - Charge usage fee $> MC$.
- What do you think happened?
 - Competitors flooded the market with knock-off k-cups
- How did the company respond?

Keurig 2.0 machines to feature RFID-limited K-cups

Will coffee drinkers be stupid enough to fall for this? Stay tuned



Jennifer Abel



Keurig competitors crack company's DRM code

This is why your old K-cups won't work in new Keurig machines



Jennifer Abel



Possible Application (3): Keurig and K-cup



- Keurig was hoping to profit off of K-cup sales, and in some cases they will. But consumers have balked at being required to buy K-cups specifically.
- Therefore, Keurig may need to raise the price of their machines
- **What would have been a smarter strategy....?**
- To have from the beginning a premium model that can take any cup, and a “bargain” model that only uses K-cups. Self-selecting pricing (“damaged goods”) meets two-part tariff.

Summary & Direction

Today

- Membership plans: When $MC=0$ and consumer valuation falls w/quantity, better off to offer single price equal to total valuation of max nr of units for which reservation price is positive
- Two-part tariffs: When MC not 0, price plans that involve a membership fee and a usage fee. In markets with a single type of consumer, optimal plan is to charge MC as the usage fee and consumer surplus as the membership fee.
- With multiple types, the membership fee must be low enough to entice light users to join, thus you set the usage fee higher to make money on heavy users.

Next Lecture

Risk aversion: how do consumers behave in the presence of risk?
Certainty equivalent: how much would I pay to avoid risk?
Adverse selection: How do consumers “sort” into products?